

1(a)	<i>idea of:</i> <i>grinding with sand</i> to break open the cells / release enzyme ; <i>filtering through cloth</i> to remove, sand / debris / AW ;	2
1(b)(i)	(<i>idea that</i> result / it, is) quantitative / AW ;	1
1(b)(ii)	<i>independent variable:</i> <u>concentration</u> of catechol / substrate ; <i>dependent variable :</i> absorbance (on colorimeter) ;	2

Question	Answer	Marks
1(c)(i)	any eight from: 1 <i>ref. to</i> preparing / using, at least five dilutions concentrations of catechol ; 2 stated range of at least five concentrations from 1% downwards with % units ; 3 method of keeping catechol oxidase / enzyme solution / banana extract cold ; 4 <i>idea of</i> bringing enzyme and substrate to temperature / equilibrating, <u>before</u> mixing ; 5 method of keeping reactants / reaction mixture, at, constant / controlled / stated / standardised, temperature ; 6 same / constant / stated / standardised / known, volume / concentration / <u>mass</u> of, Y / inhibitor ; 7 same / constant / stated / standardised / known, volume of, catechol / substrate ; 8 same / constant / stated / standardised / known, volume / concentration, of catechol oxidase / enzyme / banana extract ; 9 <i>ref. to</i> measuring absorbance / intensity of colour (on colorimeter) after <u>one</u> set time ; 10 additional detail of use of colorimeter ; 11 repeat at least twice / three replicates, <u>and</u> calculate mean ; 12 <i>idea of</i> repeating, whole experiment / range of concentrations, without, Y / inhibitor ; 13 named hazard <u>and</u> risk <u>and</u> precaution ;	8
1(c)(ii)	curve should be to the right of the original curve ; curve should plateau below the original curve ;	2
1(c)(iii)	correct placement of $V_{\max}$ <u>and</u> K_m ;	2
1(c)(iv)	(reaction with non-competitive inhibitor) has same K_m value than with no inhibitor ;	1

Question	Answer	Marks
1(d)	any three from: 1 combination of drugs gives a larger reduction in blood pressure ; 2 no error bars ; 3 no, statistical test / <i>t</i>-test, done to compare groups ; 4 no data on long term results ; 5 AVP ; ; any two from: • no information on blood pressure prior to trial • no information on type of blood pressure • no information on number of patients in each group • no data on side effects • <i>ref. to</i> reduction in blood pressure brought about by the placebo alone • no data on, age / gender	3